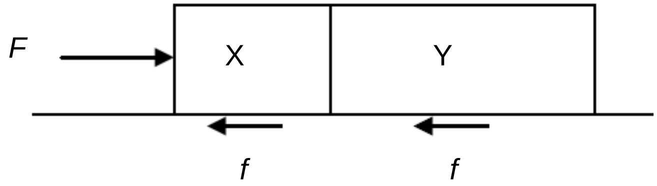
4 Two blocks, X and Y, of masses m and $2m$ respectively, are accelerated along a rough surface by a force F applied to block X. The magnitude of the frictional force experienced by each of the block is f.



What is the magnitude of the force exerted by block Y on block X?

A $\frac{1}{3}(F-2f)$

B $\frac{1}{3}(F-f)$

C $\frac{1}{3}(2F-f)$

D $\frac{2}{3}(F-f)$

Ans: C

$F_{net} = F - 2f = (m + 2m)a_{net} \rightarrow a_{net} = (F - 2f)/3m$

Analysing FBD of X:

$F - f - F_{Y\text{-}ON\text{-}X} = ma_{net} \rightarrow F_{Y\text{-}ON\text{-}X} = F - f - m[(F - 2f)/3m]$

$= F - f - F/3 + 2/3\ f = 2/3\ F - 1/3\ f = 1/3\ (2F - f)$

5 Two steel balls X and Y are suspended on strings. Ball X is pulled to one side as shown